Advanced Representation Learning
(고급표현학습)

Instructor: Donghyeon Cho

The Department of Computer Science

Hanyang University

한양대학교
HANYANG UNIVERSITY





Course Information

- Instructor
  - Prof. Donghyeon Cho
  - Office: IT.BT rm 504
  - Tel: 02-2220-2657
  - Web: cvlab.hanyang.ac.kr
  - Email: doncho@hanyang.ac.kr; cdh12242@gmail.com
  - Research area: computer vision, visual intelligence
- Smart-F (Flip Learning)
  - Online + offline (1.5h)
  - Online: basic concept
  - Offline: review + extra
    - You can do paper presentations if you want~!

Prerequisite subjects

- Deep learning
  - Not for beginner’s course
  - Dealing with “(slightly) Advanced” topics!
  - Course for students majoring in AI
- No textbook!

Time Table

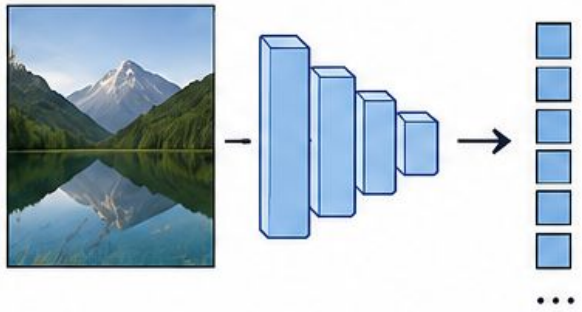
Date	Lecture	Details
Week 1 (9/3(목))	Introduction	
Week 2 (9/10(목))	No class (휴강)	ECCV Periods
Week 3 (9/17(목))	Self-Supervised Learning	
Week 4 (9/24(목))	Contrastive Learning I	Chuseok Periods (online)
Week 5 (10/1(목))	Contrastive Learning II	
Week 6 (10/8(목))	Contrastive Learning III	
Week 7 (10/15(목))	Semi-Supervised Learning I	
Week 8 (10/22(목))	Semi-Supervised Learning II	
Week 9 (10/29(목))	Midterm Quiz	
Week 10 (11/5(목))	Weakly-Supervised Learning I	
Week 11 (11/12(목))	Weakly-Supervised Learning II	
Week 12 (11/19(목))	Active Learning I	
Week 13 (11/26(목))	Active Learning II	
Week 14 (12/3(목))	Masked Autoencoder (MAE)	
Week 15 (12/10(목))	Final Quiz	
Week 16 (12/17(목))	VideoMAE+	

Grading

- Attendance: 10%
- Participation: 30%
- Mid Quiz: 30%
- Final Quiz: 30%

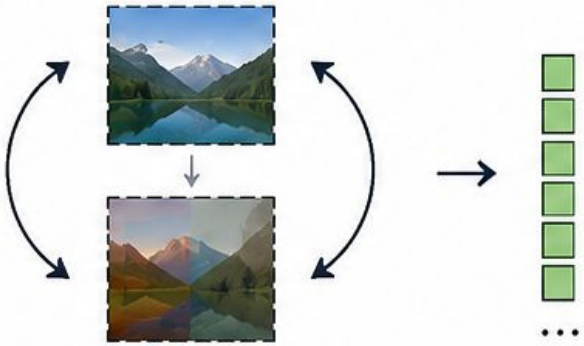
Introduction

Feature Learning



Learn useful features

Self-Supervised Learning



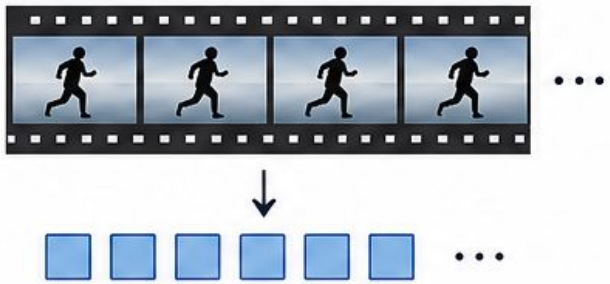
Learn from unlabeled data

Foundation Models



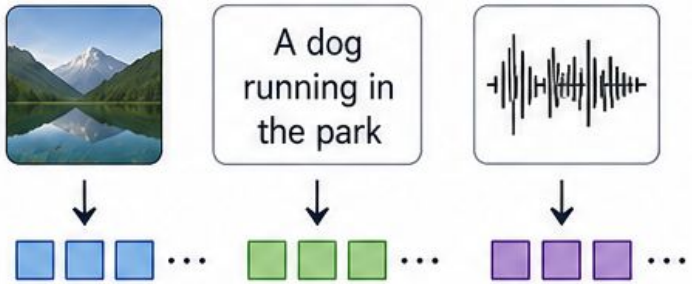
Generalize to many tasks

Video Representation Learning



Capture temporal dynamics

Multimodal Representation Learning



Align across modalities

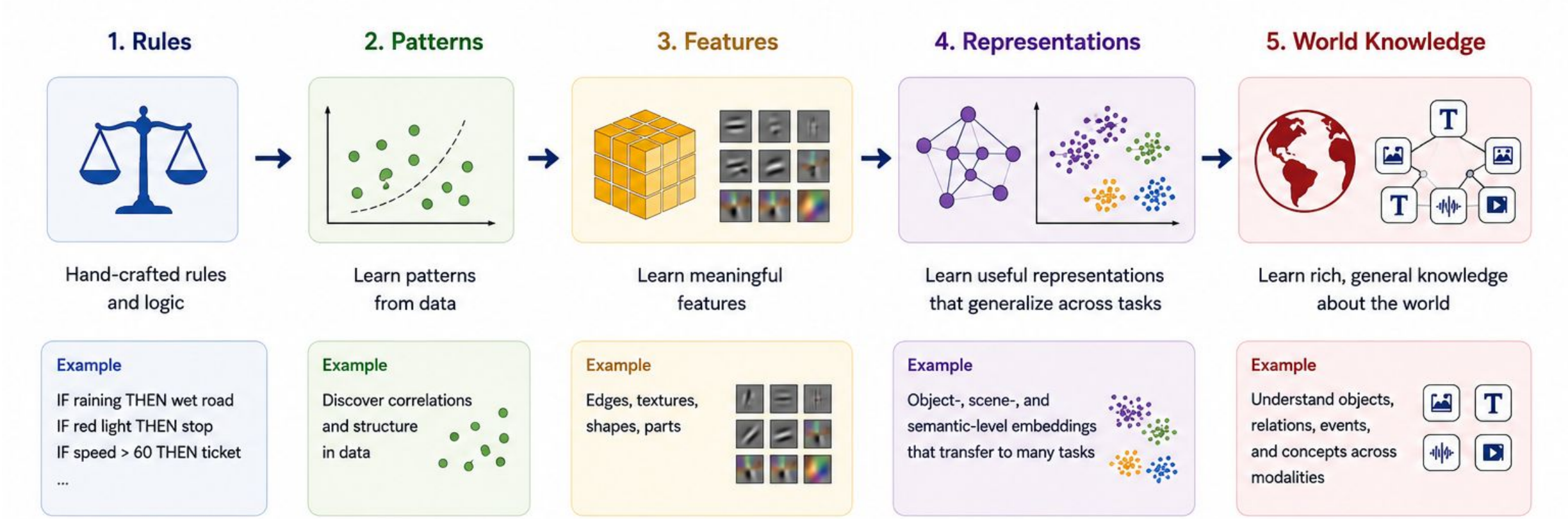
Applications

- Classification
- Detection
- Segmentation
- Generation
- ...

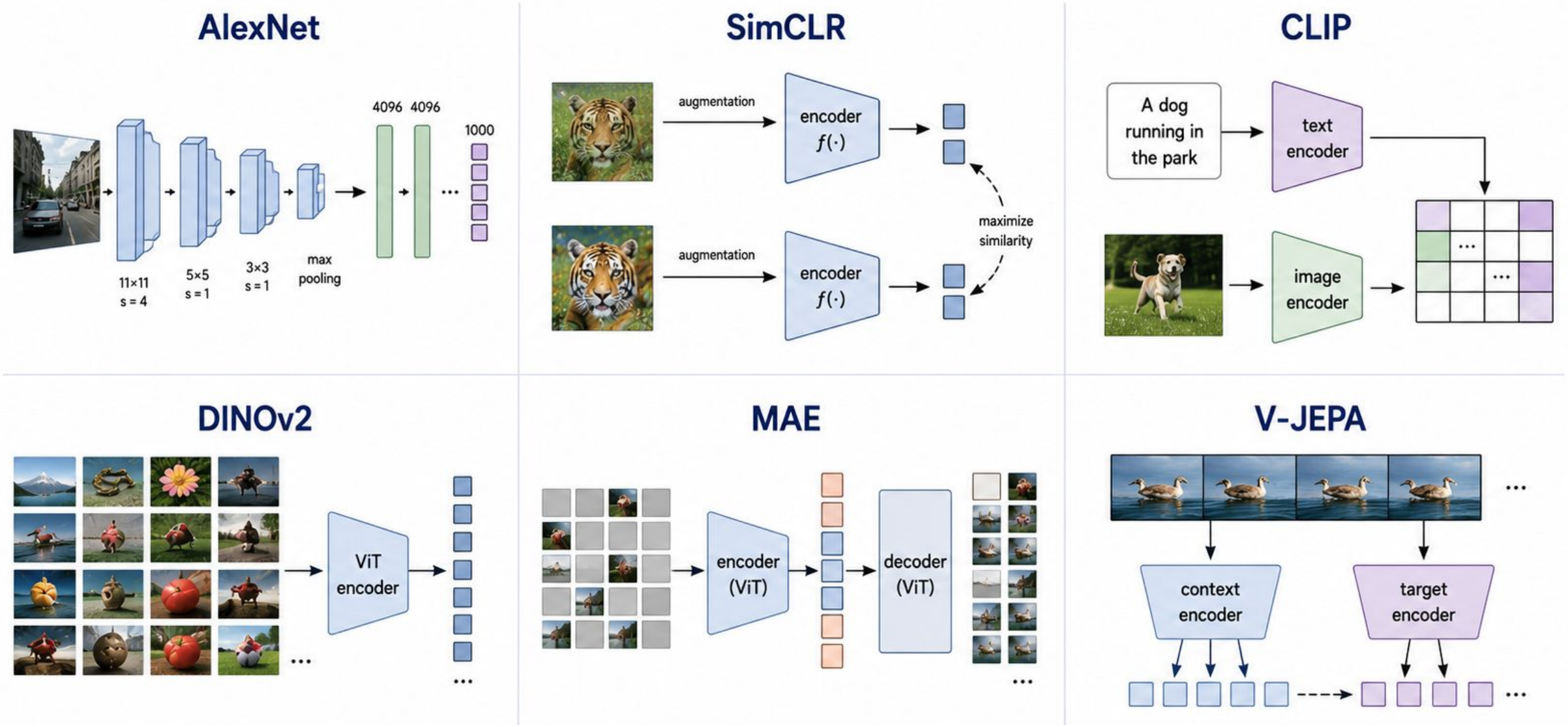
From understanding data to understanding the world.

What AI has Been Learning

- From rules to world knowledge: the journey of learning better representations



Representative Models



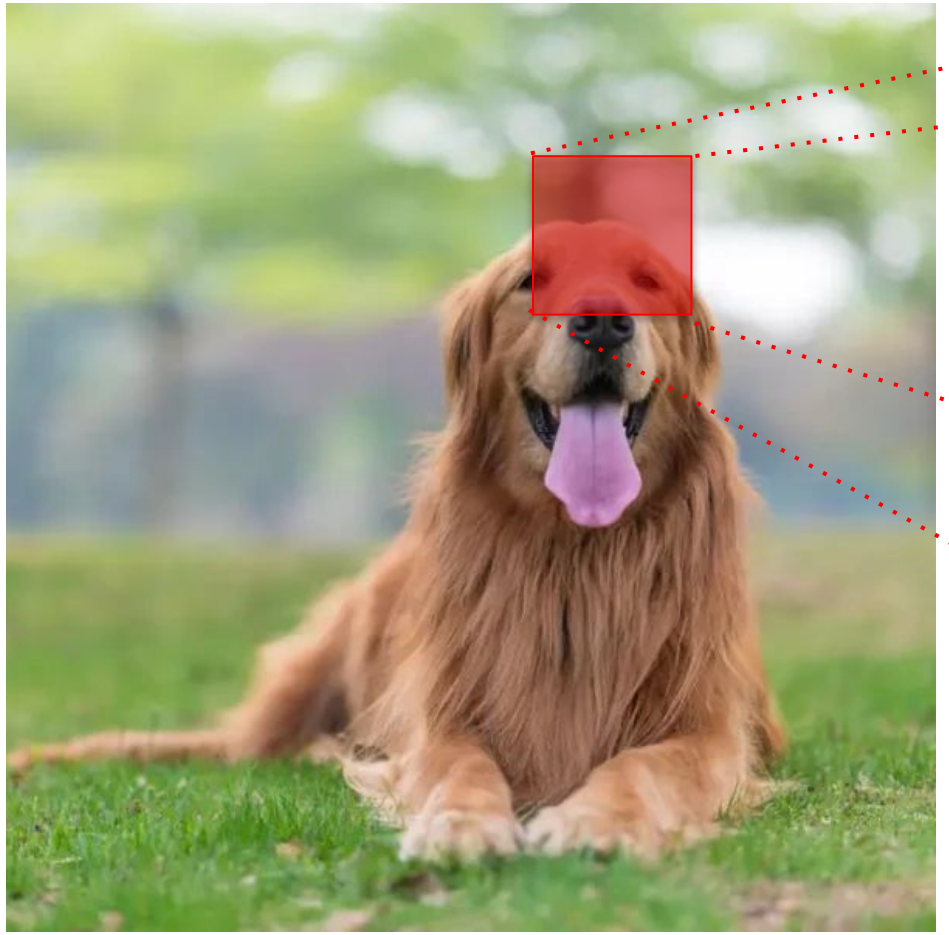
What is a Representation?

- How Does a Computer Understand an Image?



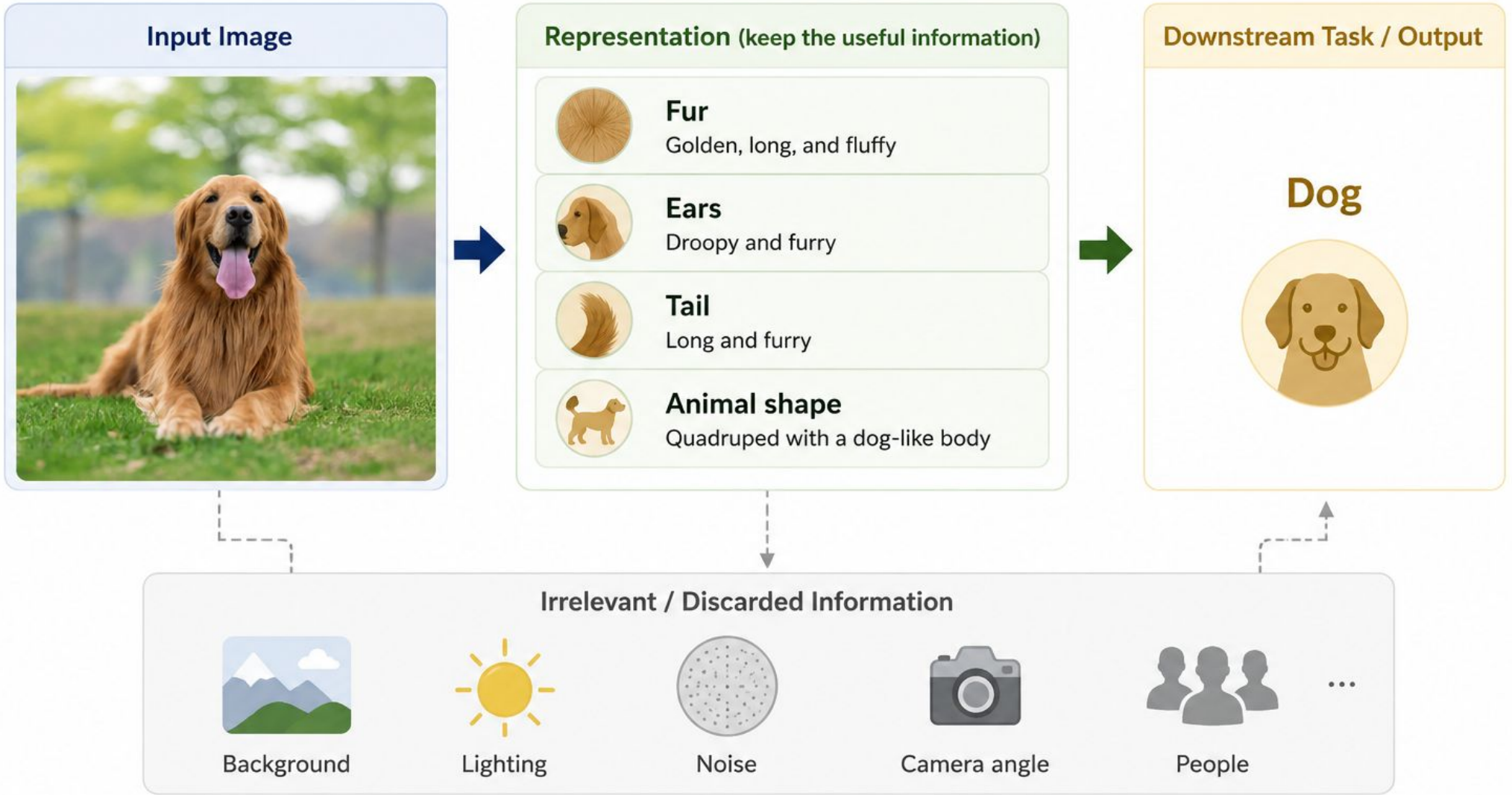
Raw Pixels Are Not Understanding

- To a computer, an image is initially just a matrix of numbers.



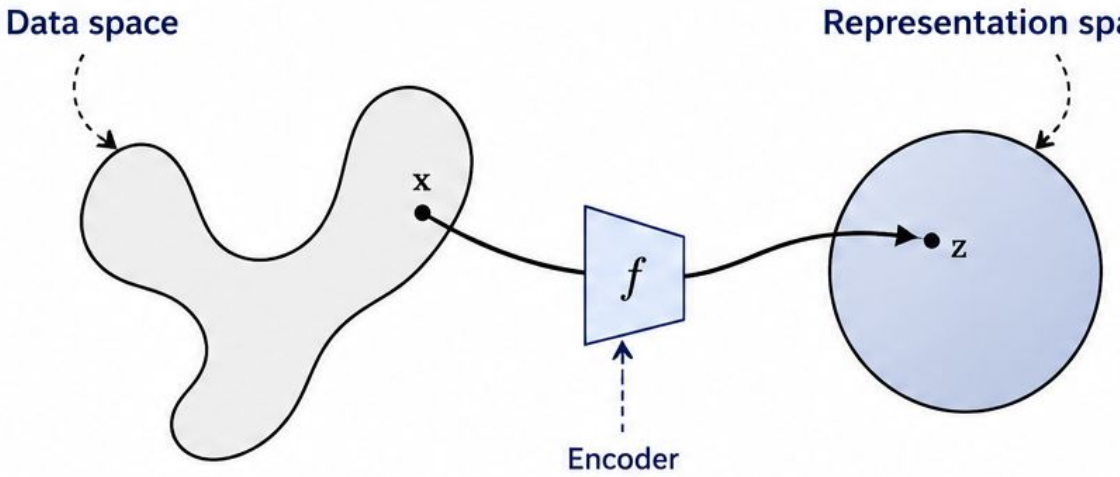
100	100	100	128	128	128	130
100	100	100	128	128	128	130
100	100	100	128	128	128	130
100	100	100	128	128	128	130
100	100	100	128	128	128	130
100	100	100	128	128	128	130
100	100	100	128	128	128	130
100	100	100	128	128	128	130
100	100	100	128	128	128	150
100	100	100	128	128	128	150

What Information Should be Kept?



Representation

- Representation learning aims to map data points to a simpler, organized space.



Mapping (Encoding)

$$\mathbf{z} = f(\mathbf{x})$$

$\mathbf{x} \in \mathcal{X}$ (data space)

$\mathbf{z} \in \mathcal{Z}$ (representation space)



Input data

$\mathbf{x} \in \mathcal{X}$

Encoder f

$$\begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_D \end{bmatrix}$$

Representation

$\mathbf{z} = f(\mathbf{x}) \in \mathcal{Z}$

Downstream task

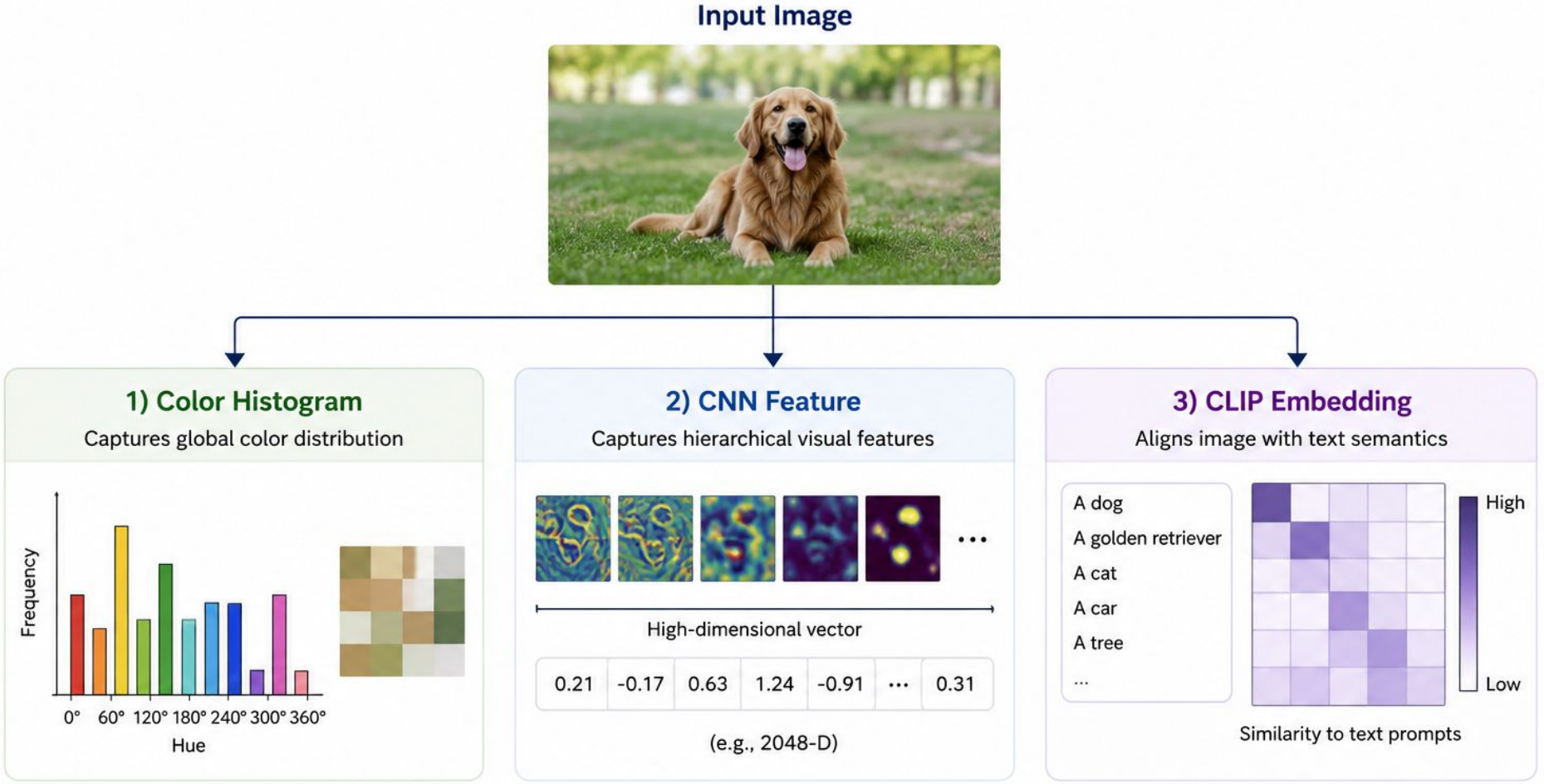
- Classification
- Detection
- Retrieval
- ...

- ✓ Learns useful structure
- ✓ Captures semantics
- ✓ Enables many tasks
- ✓ Often in lower dimension $(D \ll \dim(\mathbf{x}))$

⭐ **A representation is a transformation of raw data into a form that makes downstream tasks easier.**

Many Possible Representations

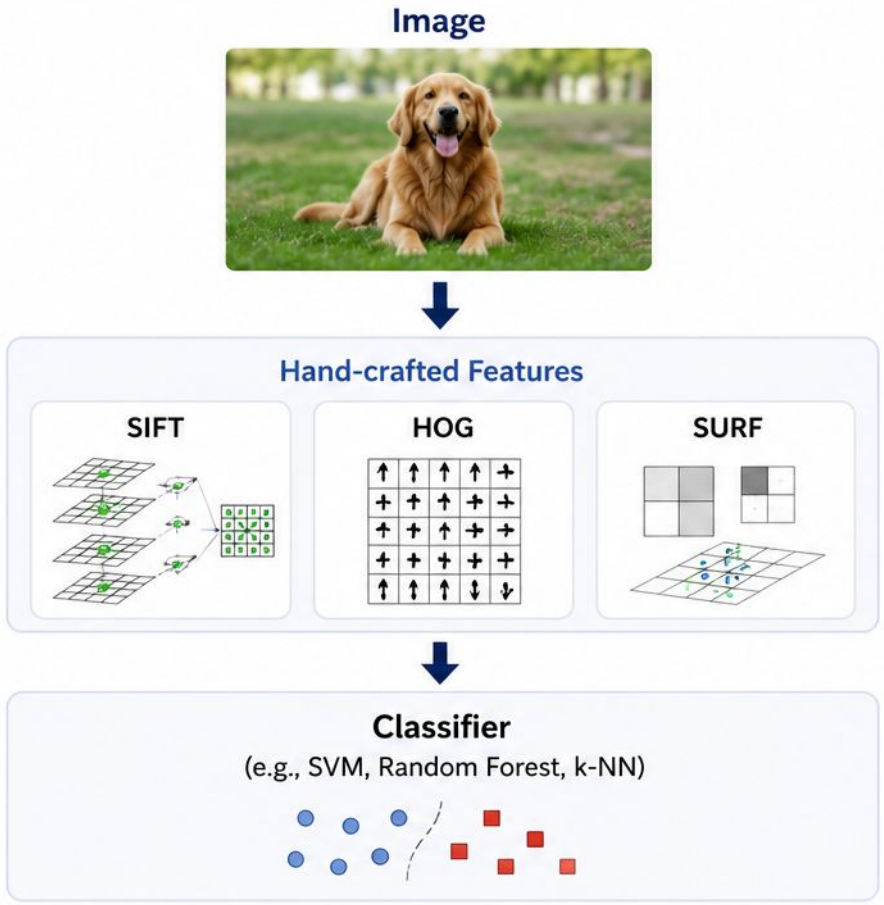
- The same data can have many different representations.

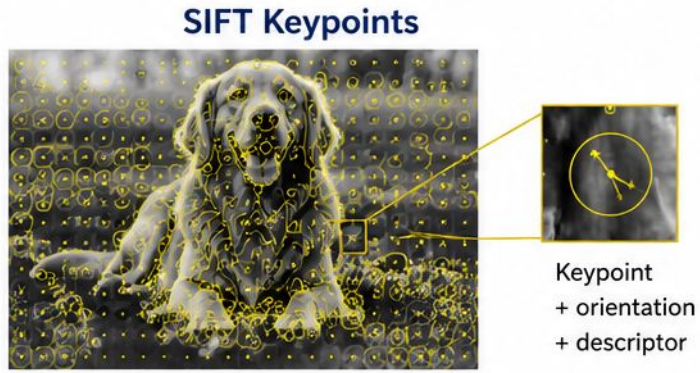


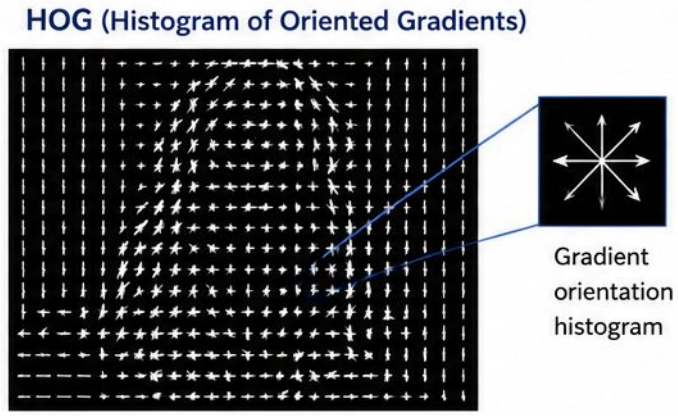
The same data can have many different representations.

Before Deep Learning

- Who design the features?







Feature Engineering Era: Humans manually designed features.

Different tasks require different features.

Why is This Problematic?

1) Human-designed

- Features are crafted by experts
- Require domain knowledge
- Time-consuming and labor-intensive

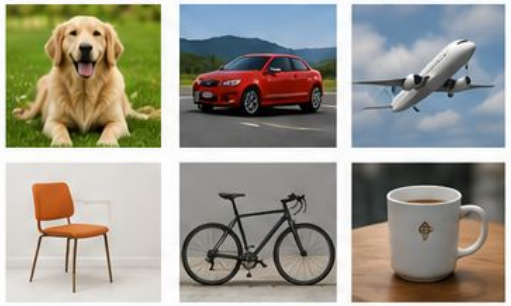
2) Task-specific

- Features optimized for one task
- Do not generalize well to others
- New task = new feature design

3) Limited scalability

- Does not scale to complex data
- Hard to cover all variations
- Limited performance ceiling

Object Recognition



Requires specific features for objects and shapes

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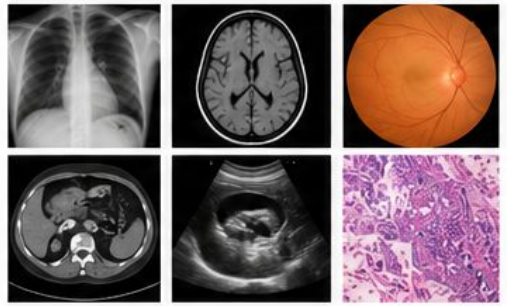
Face Recognition



Requires specific features for facial patterns

≠

Medical Imaging



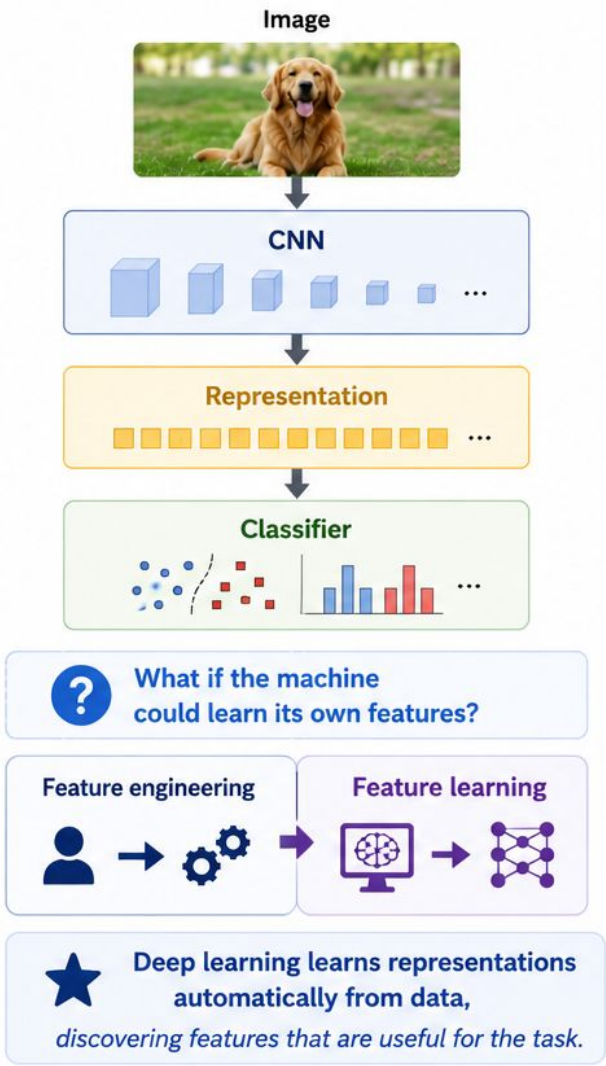
Requires specific features for medical patterns

Every new task requires new features.

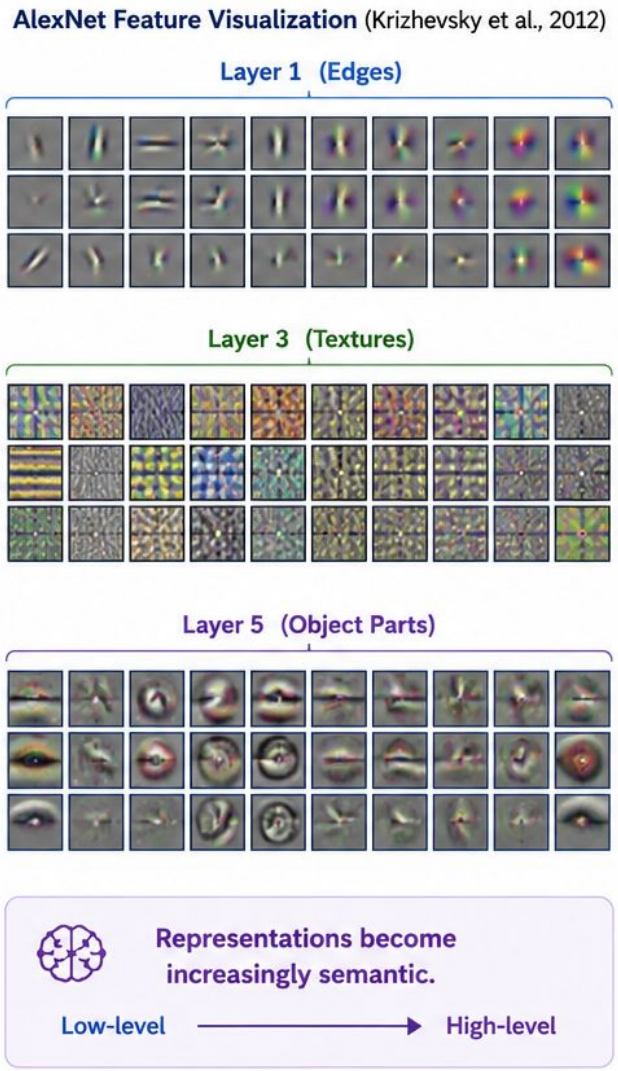
This limits generalization, efficiency, and performance.

Deep Learning Changes the Game

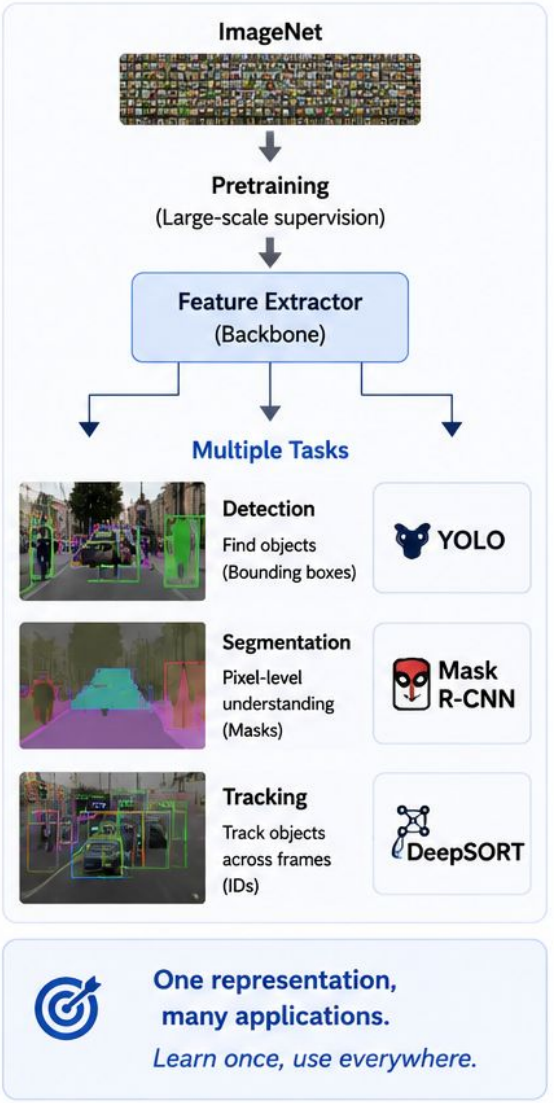
1. From Hand-Crafted to Learned Representations



2. What Does CNN Learn?

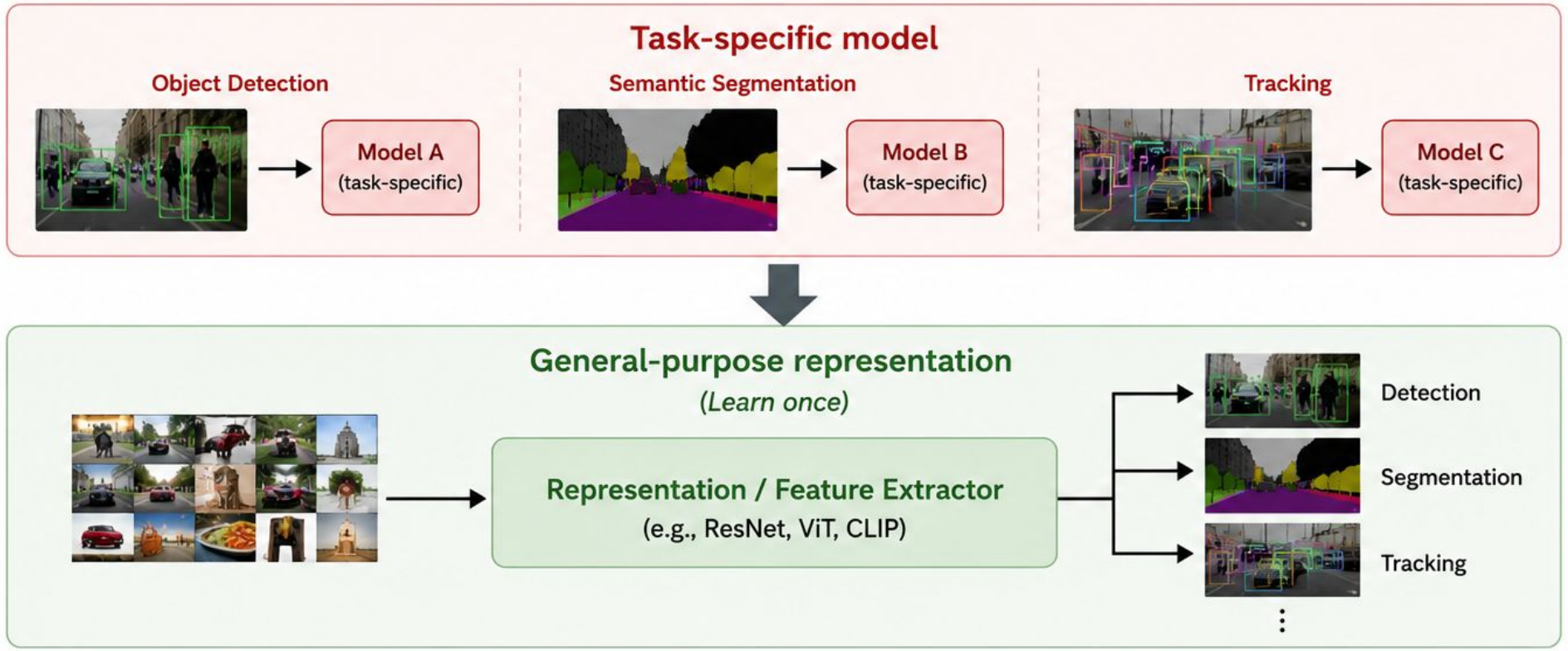


3. Representation Reuse



What is the Good Representations?

- A good representation can be reused across many tasks.



What Makes a Good Representation?

- A good representation should:
  - Compress information
  - Support prediction
  - Generalize across tasks

Compression

- Captures just the essential characteristics of the data.
  - Less Memory
    - Smaller representations are easier to store.
  - Invariance
    - Remove nuisance factors that do not affect semantics.
  - Occam's Razor
    - Among equally useful representations, prefer the simpler one.

Compression is not about losing information.
It is about removing irrelevant information.

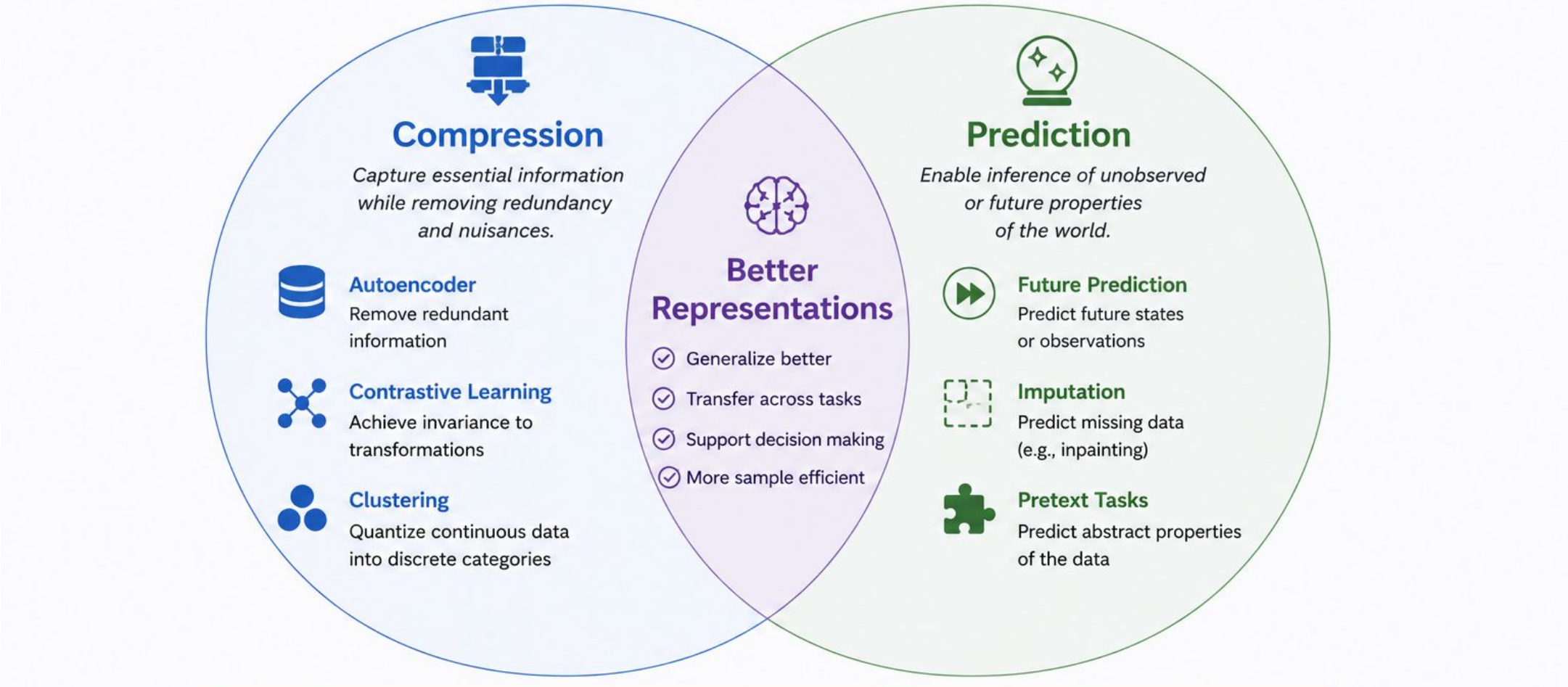
Prediction

- Why do we need representations?
  - To predict what we cannot directly observe.



Prediction means inferring unobserved properties of the world.

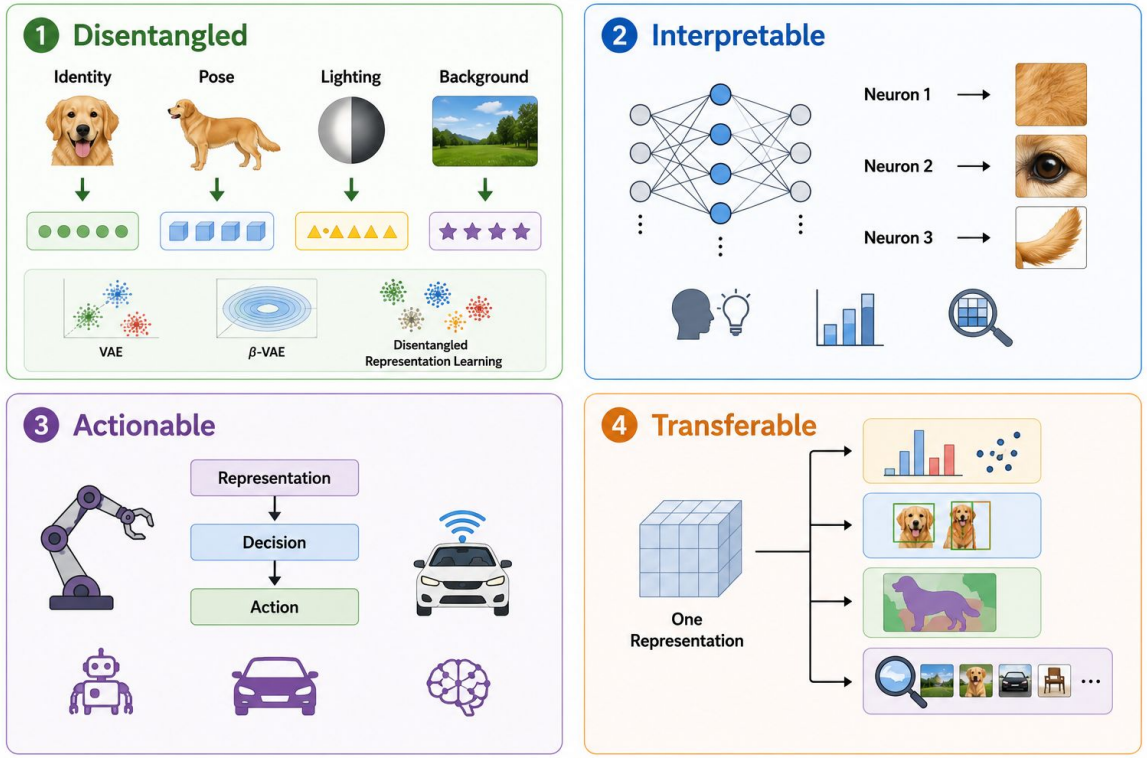
Good Representations = Compression + Prediction



Beyond Compression and Prediction

- Good representations often exhibit additional desirable properties.

1. Disentangled
2. Interpretable
3. Actionable
4. Transferable



These properties are often consequences of learning representations that are both

Types of Representation Learners

Learning Method	Learning Principle	Short Summary
Autoencoding	Compression	Remove redundant information
Contrastive	Compression	Achieve invariance to viewing transformations
Clustering	Compression	Quantize continuous data into discrete categories
Future prediction	Prediction	Predict the future
Imputation	Prediction	Predict missing data
Pretext tasks	Prediction	Predict abstract properties of your data

Next Week..

- Self-Supervised Learning and Pretext Tasks